AI-Powered Driving Efficiency Optimizer

MATLAB CODE

```
clc;
clear;
close all;
%% Step 1: Generate Simulated Driving Data
N = 500;   % Number of samples
speed = randi([20 120],N,1);             % km/h
acceleration = -3 + 6*rand(N,1);         % m/s^2
gradient = -5 + 15*rand(N,1);            % Road slope (degrees)
brakeCount = randi([0 15],N,1);          % Number of braking events
temperature = randi([15 40],N,1);        % Celsius
%% Step 2: Calculate Energy Consumption (Target)
energy = 120 ...
    + 0.6*speed ...
    + 10*abs(acceleration) ...
    + 3*gradient ...
    + 2*brakeCount ...
    - 0.5*temperature ...
    + randn(N,1)*5;
%% Step 3: Create Input Matrix
X = [speed acceleration gradient brakeCount temperature];
Y = energy;
%% Step 4: Split Dataset
cv = cvpartition(N,'HoldOut',0.20);
Xtrain = X(training(cv),:);
Ytrain = Y(training(cv));
Xtest = X(test(cv),:);
Ytest = Y(test(cv));
%% Step 5: Train AI Model
model = fitrensemble(Xtrain,Ytrain);
```

```
% Step 6: Predict Energy Consumption

Ypred = predict(model,Xtest);

%% Step 7: Calculate Accuracy

RMSE = sqrt(mean((Ypred-Ytest).^2))

fprintf('\n====================================\n');

fprintf(' AI MODEL RESULTS\n');

fprintf('====================================\n');

fprintf('Root Mean Square Error (RMSE): %.2f Wh/km\n',RMSE);

%% Step 8: Plot Results

figure;

plot(Ytest,'b','LineWidth',2)

hold on

plot(Ypred,'r--','LineWidth',2)

xlabel('Test Sample')

ylabel('Energy Consumption (Wh/km)')

title('Actual vs Predicted Energy Consumption')

legend('Actual','Predicted')

grid on

%% Step 9: Predict for New Driver

disp(' ')

disp('====================================')

disp('NEW DRIVER ANALYSIS')

disp('====================================')

newSpeed = 95;

newAcceleration = 2.5;

newGradient = 4;

newBrake = 10;

newTemperature = 30;

newInput = [newSpeed newAcceleration newGradient newBrake newTemperature];

predictedEnergy = predict(model,newInput);

fprintf('Predicted Energy Consumption = %.2f Wh/km\n',predictedEnergy);
```

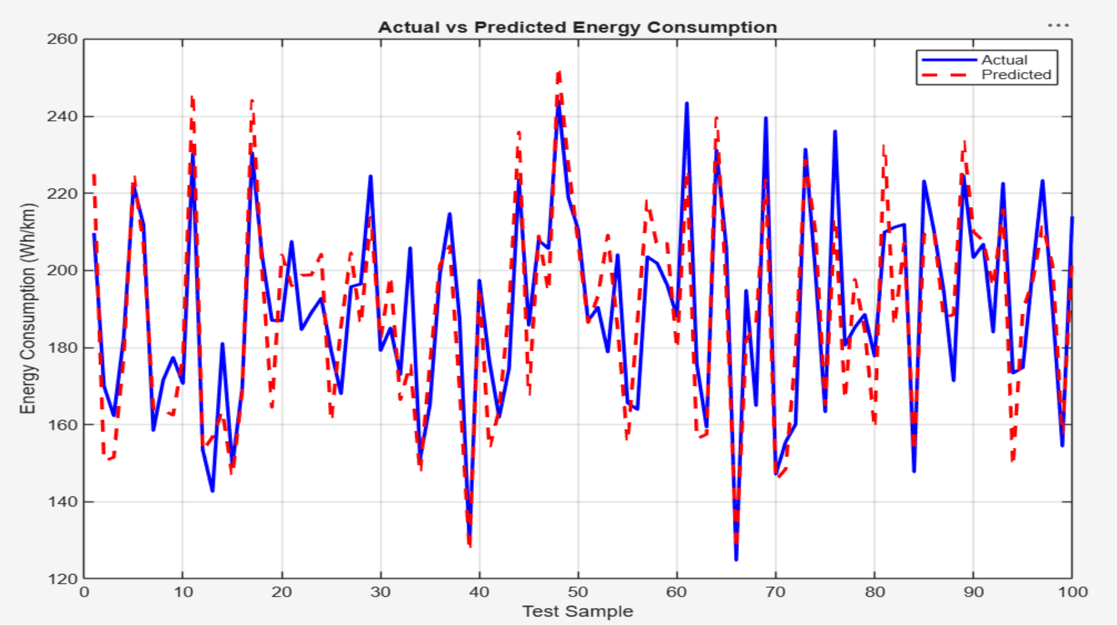
```
%% Step 10: Driving Suggestions
disp(' ')
disp('Driving Suggestions')
if newSpeed > 80
   disp('• Reduce speed below 80 km/h to improve efficiency.')
end
if abs(newAcceleration) > 2
   disp('• Avoid sudden acceleration.')
end
if newBrake > 8
   disp('• Brake smoothly and use regenerative braking.')
end
if newGradient > 5
   disp('• Steep roads increase energy usage.')
end
if newTemperature > 35
   disp('• High temperature may increase battery cooling load.')
end
if newSpeed <= 80 && abs(newAcceleration)<=2 && newBrake<=8
   disp('• Excellent driving style! Energy usage is optimized.')
end
%% Step 11: Feature Importance
figure
predictorImportance = predictorImportance(model);
bar(predictorImportance)
set(gca,'XTickLabel',{'Speed','Acceleration','Gradient','Brake','Temperature'})
xlabel('Driving Parameters')
ylabel('Importance')
title('Feature Importance')
grid on
```

```
%% Step 12: Summary

disp(' ')

disp('====================================')

disp('PROJECT SUMMARY')

disp('====================================')

fprintf('Dataset Size        : %d Samples\n',N);

fprintf('Training Samples     : %d\n',length(Ytrain));

fprintf('Testing Samples      : %d\n',length(Ytest));

fprintf('Prediction RMSE      : %.2f Wh/km\n',RMSE);

fprintf('Predicted Energy     : %.2f Wh/km\n',predictedEnergy);

disp('Project Completed Successfully!')
```

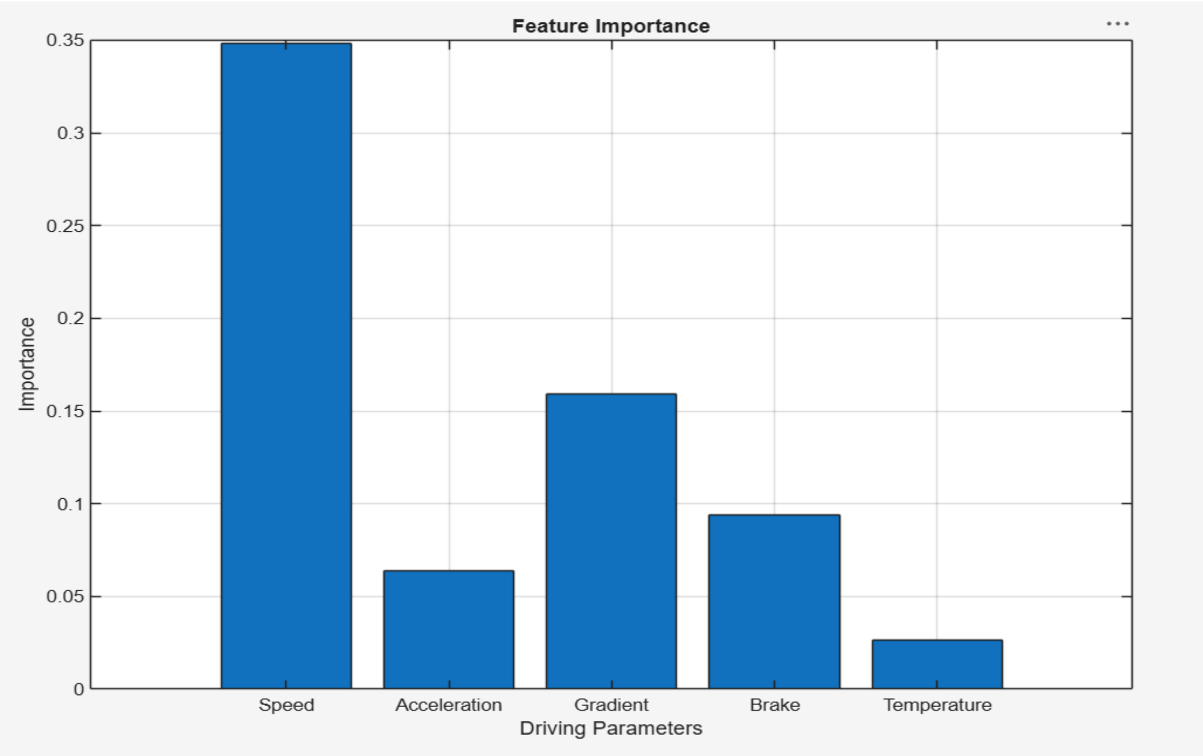
OUTPUT:-

Command Window

```
=====================================
 AI MODEL RESULTS
=====================================
Root Mean Square Error (RMSE): 12.97 Wh/km

=====================================
NEW DRIVER ANALYSIS
=====================================
Predicted Energy Consumption = 217.25 Wh/km

Driving Suggestions
• Reduce speed below 80 km/h to improve efficiency.
• Avoid sudden acceleration.
• Brake smoothly and use regenerative braking.

=====================================
PROJECT SUMMARY
=====================================
Dataset Size          : 500 Samples
Training Samples      : 400
Testing Samples       : 100
Prediction RMSE       : 12.97 Wh/km
Predicted Energy      : 217.25 Wh/km
Project Completed Successfully!
>>
```



Actual vs Predicted Energy Consumption
Actual
Predicted
Energy Consumption (Wh/km)
Test Sample
260
240
220
200
180
160
140
120
0
10
20
30
40
50
60
70
80
90
100



Feature Importance
Importance
Driving Parameters
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
Speed
Acceleration
Gradient
Brake
Temperature